The Arab American University

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Linux Admin Dashboard Project Report

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1. Introduction

This project is a Linux-based terminal program made using Bash scripting.
It provides a menu system that helps users perform basic system administration tasks in Linux.

The project runs in the terminal and helps manage system information, users, files, logs, backups, and reports.

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2. Project Objective

The main goals of this project are:

- Learn Linux commands
- Practice Bash scripting
- Build a simple admin tool
- Automate basic system tasks
- Understand Linux file and user management

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3 Tools Used

- Linux Operating System (Ubuntu)
- Bash Shell Scripting
- Terminal (Command Line)
- Basic Linux commands (find, grep, tar, etc.)

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4. Project Description

This project is a menu-based system.
The user selects an option from the menu, and the system performs the task.

The dashboard includes the following modules:

4.1 System Information

This section shows basic system details like:

- Hostname
- Current user
- System uptime
- Memory usage
- Disk usage

4.2 Users and Groups

This section manages user information:

- Show all system users
- Show current user groups
- Show logged-in users
- Count total users

4.3 File Management

This section helps manage files:

- Search for files
- Count files in a directory
- Show directory size
- Create new files
- Delete files

4.4 Permission Audit

This section checks file permissions:

- View file permissions
- Find files with full permissions (777)

4.5 Log Analysis

This section reads system logs:

- Show last system log entries
- Search logs by keyword

4.6 Backup System

This section creates backups:

- Compress selected directories
- Save backup as .tar.gz file

4.7 Report Generation

This section creates a system report file that includes:

- Date
- User
- Hostname
- Uptime
- Memory usage
- Disk usage

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5. Sample Input and Output

Example 1: System Information

Input:

```
1
```

Output:

```
Hostname: suhai-PC
Current User: suhai
Uptime: 2 hours
Memory Usage: ...
Disk Usage: ...
```

Example 2: File Search

Input:

```
3 → 1
linux_dashboard.sh
```

Output:

```
./linux_dashboard.sh
```

Example 3: Backup System

Input:

```
6 → 1
```

Output:

```
Backup created successfully:
backup_20260612_153000.tar.gz
```

Example 4: Report Generation

Input:

```
7
```

Output:

```
system_report.txt created successfully
```

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6. Linux Commands Used

- `hostname` → shows system name
- `whoami` → shows current user
- `uptime` → shows system running time
- `free -h` → shows memory usage
- `df -h` → shows disk usage
- `find` → searches files
- `grep` → searches text inside files
- `cut` → extracts text columns
- `wc -l` → counts lines
- `tar` → creates backup files
- `chmod` → changes file permissions
- `who` → shows logged-in users

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7. Conclusion

This project helped improve understanding of Linux systems and Bash scripting.
It shows how to automate system administration tasks using simple scripts.

The project also demonstrates real Linux commands used in daily system management.